The Single-Generator Theory of Cycle-Decomposed Digit-Subtraction Dynamics

A structural approach to periodicity and cycle decomposition

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Abstract

For a base $B \geq 2$, digit length k, and permutation $\pi \in S_k$, we study the arithmetic dynamical system F_π obtained by subtracting each local base-B numeral along the disjoint cycles of π from its cyclic rotation and scattering the resulting digits back to their original positions. We establish a complete closed-form theory for a single generator π, uniformly across all bases and cycle types. The cycle structure yields an exact direct-product decomposition of F_π into local maps g_t, and each g_t is semiconjugate, after identifying $x \sim -x$, to multiplication by

$$c = (B-1)B^{t-1}$$

on $\mathbb{Z}/(B^t-1)\mathbb{Z}$. Writing

$$B^t - 1 = N_1 N_2,$$

where every prime divisor of N_1 divides $B-1$ and $\gcd(N_2, B-1) = 1$, this arithmetic reduction determines the exact fixed-point count, the criterion for total collapse, and the complete attracting-cycle structure. Each local map has $(N_2+1)/2$ periodic states. Their individual periods are determined by the divisor structure of N_2: for a periodic class represented by $N_1 r$, the minimal period is the least positive integer j such that $c^j \equiv \pm 1$ modulo $N_2/\gcd(r, N_2)$. The quantity $P_* := \mathrm{ord}_{(\mathbb{Z}/N_2\mathbb{Z})^\times/\{\pm 1\}}(c)$ is the least common multiple of all periodic-orbit lengths. Every state reaches the periodic set after exactly a_0 or a_0+1 iterations, where a_0 is the least integer satisfying $N_1 \mid c^{a_0}$. The direct-product decomposition then yields a closed formula for the eventual rank of F_π for an arbitrary cycle type. For a two-cycle, this rank is $(\mathrm{odd}(B+1)+1)/2$, giving the identification with OEIS A003602 as a direct consequence of the underlying 2-adic decomposition.

1. Introduction

Iterated digit-rearrangement-and-subtraction maps have been studied since Kaprekar observed that sorting the digits of a four-digit base-ten integer into descending and ascending order and subtracting drives every non-repdigit to the fixed point 6174 within seven steps [3, 4]. Writing $T_{b,n}$ for the analogous transformation on n-digit base-b integers — sort descending, sort ascending, subtract, iterate — the resulting theory of fixed points and cycles is by now substantial and, in odd bases at least, largely complete. Yamagami and Matsui [11] gave a full classification for $b = 3$ in terms of subgroups and cosets of the multiplicative group $(\mathbb{Z}/(n-2)\mathbb{Z})^\times$; Yamagami [10] resolved $b = 2$ completely; and Kay and Downes-Ward [2] extended the theory beyond $b = 3$ to all odd bases, uncovering qualitatively different behaviour

according to whether $b \equiv 1$ or $b \equiv 3 \pmod 4$, an extensive family of "special" cycles falling outside the main classification, and several explicit conjectures that remain open even for $b = 5, 7$. Earlier work of Lapenta, Ludington, and Prichett [6, 8] established general succession formulae and enumerated Kaprekar constants across bases. A separate generalization, due to Smarandache [9], replaces the descending/ascending sort with a pair of arbitrary digit permutations P_1, P_2 and studies $|P_1(x) - P_2(x)|$.

The construction differs from the classical routines in one essential respect: the permutation is applied cyclewise. Each cycle contributes an independent local subtraction, and the resulting digits are scattered back to the same positions. This gives a direct-product decomposition of F_π and reduces the local problem to a map on t digits.

The local map is semiconjugate to multiplication by

$$c = (B-1)B^{t-1}$$

modulo $B^t - 1$ (Theorem 3.3). This reduction yields the fixed-point count, the collapse criterion, the periodic structure, and the eventual rank. In particular, the periods are multiplicative orders modulo divisors of the part of $B^t - 1$ coprime to $B - 1$. The quantity $P_* := \mathrm{ord}_{(\mathbb{Z}/N_2\mathbb{Z})^\times/\{\pm 1\}}(c)$ is the least common multiple of the individual periodic-orbit lengths; for $B = 14$ and $t = 3$, this gives $P_* = 105$.

We restrict throughout to a single generator. Compositions of generators with different cycle structures are discussed in §8.

1.1. The Construction

Fix a base $B \geq 2$ and a digit-length $k \geq 1$, and identify a k-digit base-B numeral with its digit vector $d = (d_1, \ldots, d_k) \in \{0, \ldots, B-1\}^k$, d_1 most significant. For $\pi \in S_k$ with disjoint cycle decomposition $\pi = \pi_1 \pi_2 \cdots \pi_m$, we define a map F_π on digit vectors by acting independently on each cycle, as follows; we fix once and for all the normalization below.

Convention 1.1 (Canonical cycle form)**.** Each cycle of π is written starting from its smallest element: the cycle on $\{1, 3\}$ is written $(1\,3)$, never $(3\,1)$; the cycle on $\{4, 5, 6\}$ is written $(4\,5\,6)$.

For a cycle $\pi_i = (a_1\, a_2\, \cdots\, a_{t_i})$ in canonical form (a_1 minimal, and $a_1 \to a_2 \to \cdots \to a_{t_i} \to a_1$ under π), define:

(1) *Read.* Form the local t_i-digit number x_i from the digits at positions $a_1, \ldots, a_{t_i}$, in that order.

(2) *Rotate.* Since π_i sends $a_j \mapsto a_{j+1}$ (indices mod t_i), the digit at position a_j moves to position a_{j+1}; reading off the result in the order $a_1, \ldots, a_{t_i}$ gives $\rho_{t_i}(x_i)$, the cyclic right-rotation of x_i.

(3) *Subtract.* Set $g_i = |x_i - \rho_{t_i}(x_i)|$, a t_i-digit string $(r_1, \ldots, r_{t_i})$ (leading zeros retained).

(4) *Scatter.* Write r_j back to position a_j — the position it was read *from* in step (1), not into a contiguous output block.

Carrying this out on every cycle of π defines $F_\pi \colon \{0, \ldots, B^k - 1\} \to \{0, \ldots, B^k - 1\}$.

Example 1.2. For $B = 10$, $k = 4$, $\pi = (1\,3)(2\,4)$, $x = 5678$: the cycle $(1\,3)$ reads $x_1 = 57$, rotates to 75, and yields $|57 - 75| = 18 = (1, 8)$, scattered so that position 1 receives 1 and position 3 receives 8; the cycle $(2\,4)$ reads $x_2 = 68$, rotates to 86, yields $|68 - 86| = 18 = (1, 8)$, scattered so that position 2 receives 1 and position 4 receives 8. Hence $F_\pi(5678) = 1188$.

The map is not $|x - \pi(x)|$: in this example, $|5678 - 7856| = 2178 \neq 1188$. The cyclewise subtraction is essential to the product decomposition of §2.

We fix the canonical representative of each cycle. Other choices give conjugate maps, as shown in Proposition 1.3.

Proposition 1.3. *Let F_π and F'_π be the maps obtained from two admissible choices of starting element for each cycle of π. Then F_π and F'_π are conjugate: there is a bijection of digit vectors, matching vectors with equal local values on every cycle, intertwining the two maps.*

Proof. Fix a cycle of π of length t and two admissible starting elements a, a' for it. Reading the local number starting from a is, on the digit vectors supported on that cycle, a bijection ρ_a onto $\{0, \dots, B^t - 1\}$: it simply relabels the t coordinates by the order in which $a, \pi(a), \pi^2(a), \dots$ visits them, and this relabelling has an obvious inverse (write the digits of a t-digit number back onto the positions in that same order). The same holds for a', giving a second bijection $\rho_{a'}$. Because reading and scattering use the same bijection on the input and output sides (this is made precise in the proof of Theorem 2.1 below), the restriction of F_π to this cycle's coordinates is exactly $\rho_a^{-1} \circ g_t \circ \rho_a$, and likewise $\rho_{a'}^{-1} \circ g_t \circ \rho_{a'}$ for F'_π, where g_t is the local map of §2 — the same map g_t in both cases, since g_t depends only on the cycle length t and not on which element was chosen to start reading it. Composing cycle by cycle, $\varphi := \prod_i \rho_{a'_i}^{-1} \circ \rho_{a_i}$ is a bijection of digit vectors with $F'_\pi = \varphi \circ F_\pi \circ \varphi^{-1}$, i.e. $F'_\pi \circ \varphi = \varphi \circ F_\pi$. Unwinding the definition of φ, it matches a digit vector d with the unique digit vector e agreeing with d's local value on every cycle, once local values are associated with a'_i instead of a_i — exactly the correspondence claimed. □

1.2. Outline

The product decomposition is established in §2. The local arithmetic is developed in §3–§6, and the eventual rank for arbitrary cycle types is obtained in §7. The final section records the limitation to a single generator.

2. Product Decomposition

Throughout this section π is a fixed permutation of $\{1, \dots, k\}$ with cycles of lengths $t_1, \dots, t_m$, $\sum t_i = k$, in the canonical form of Convention 1.1.

Theorem 2.1. *As a map on digit vectors, F_π is the direct product, one factor per cycle of π, of the local maps g_i. That is: if two digit vectors d, e agree on every position of some cycle of π, then $F_\pi(d)$ and $F_\pi(e)$ agree on those same positions, and the common value depends only on the local number associated with that cycle.*

Proof. Fix a cycle $\pi_i = (a_1 \cdots a_t)$ of π, in canonical form, and let $j \in \{1, \dots, t\}$. Let ρ_i denote the bijection which sends the digit tuple on the cycle $(a_1, \dots, a_t)$ to the corresponding integer in $\{0, \dots, B^t - 1\}$,

$$\rho_i(c_1, \dots, c_t) = \sum_{l=1}^{t} c_l B^{t-l},$$

and let ρ_i^{-1} be its inverse, which returns the t base-B digits, with leading zeros retained. If Φ_i denotes the restriction of F_π to these coordinates, then

$$\Phi_i = \rho_i^{-1} \circ g_t \circ \rho_i.$$

Thus the restriction of F_π to a cycle depends only on the digits on that cycle, and its local dynamics is conjugate to g_t. Since this holds independently for every cycle, F_π is the direct product of the maps g_{t_i}. □

Theorem 2.2. *Let d be a digit vector, and for each cycle i of π let ℓ_i denote the (eventual) period of the local value associated with that cycle, under iteration of the local map g_i. Then the period of d under iteration of F_π is*

$$\ell(d) \;=\; \operatorname{lcm}(\ell_1, \ldots, \ell_m).$$

Proof. By Theorem 2.1, the coordinate change $d \mapsto (x_1, \ldots, x_m)$ intertwines F_π with $g_1 \times \cdots \times g_m$. Hence, for every n,

$$F_\pi^n(d) = d \quad \Longleftrightarrow \quad g_i^n(x_i) = x_i \text{ for every } i.$$

Thus n must be a common multiple of the minimal periods $\ell_1, \ldots, \ell_m$, and the least such n is $\operatorname{lcm}(\ell_1, \ldots, \ell_m)$. □

The periodic-orbit structure of F_π therefore depends on π only through its cycle type. Theorem 2.2 reduces the global dynamics to the local maps $g_t \colon \{0, \ldots, B^t - 1\} \to \{0, \ldots, B^t - 1\}$.

We will also use the following divisibility invariant.

Proposition 2.3. *For every $B \geq 2$, every k, every $\pi \in S_k$, and every digit vector x, we have $(B-1) \mid F_\pi(x)$.*

Proof. The cyclic rotation ρ_t of a digit string preserves its value modulo $B-1$, since the value of a base-B numeral modulo $B-1$ is its digit sum (as $B \equiv 1 \pmod{B-1}$), and a rotation only permutes the digits, leaving the digit sum unchanged. Hence each local result $g_i = |x_i - \rho_{t_i}(x_i)|$ is a difference of two numbers with the same digit sum mod $B-1$, so $g_i \equiv 0 \pmod{B-1}$. The digit sum written back across cycle i equals the digit sum of g_i, hence is $\equiv 0 \pmod{B-1}$; summing over all cycles, the digit sum of $F_\pi(x)$ is $\equiv 0 \pmod{B-1}$, which is exactly $F_\pi(x) \equiv 0 \pmod{B-1}$. □

3. Linearization of the Local Map

Write $M_t = B^t - 1$ and let $R_m = 1 + B + \cdots + B^{m-1}$ denote the base-B repunit of length m (so $R_t = (B^t - 1)/(B-1)$); the defining recursion $R_m = B\,R_{m-1} + 1$ (with $R_0 = 0$) is used repeatedly below. Let $\rho_t(x)$ denote the right cyclic rotation of the t-digit base-B expansion of x, with leading zeros retained.

Lemma 3.1. *For x with base-B representation $x = Bq + c$ ($c = x \bmod B$, $q = \lfloor x/B \rfloor$),*

$$g_t(x) \;=\; (B-1)\,\big|\, q - R_{t-1}\, c \,\big|.$$

Proof. Right cyclic rotation of the t-digit string $x = (c_1, \ldots, c_t)$ (with $c_t = c$) moves the last digit to the front, and one checks directly that this is the map $\rho_t(x) = q + cB^{t-1}$: dropping the last digit and shifting the rest down by one place is division by B, and placing c in the leading slot contributes cB^{t-1}. Hence

$$\begin{aligned} x - \rho_t(x) &= (Bq + c) - (q + cB^{t-1}) \\ &= (B-1)q - (B^{t-1} - 1)c \\ &= (B-1)(q - R_{t-1}c). \end{aligned}$$

using $B^{t-1} - 1 = (B-1)R_{t-1}$. Taking absolute values gives $g_t(x) = |x - \rho_t(x)| = (B-1)|q - R_{t-1}c|$, since $B - 1 \geq 0$. □

The formula also gives the image of g_t. Every value $v \in \{0, \ldots, B^{t-1} - 1\}$ occurs in $|q - R_{t-1}c|$ by taking $c = 0$ and $q = v$. Hence

$$\operatorname{Im}(g_t) = \{(B-1)v : 0 \leq v < B^{t-1}\},$$

which has cardinality B^{t-1}.

Lemma 3.2 (Complementation symmetry)**.** *$g_t(B^t - 1 - x) = g_t(x)$ for every $x < B^t$.*

Proof. Write $x = Bq + c$ as above, with $0 \le c < B$ and $0 \le q < B^{t-1}$. Since $B^t - 1 = B(B^{t-1} - 1) + (B - 1)$, we have

$$B^t - 1 - x = B\big(B^{t-1} - 1 - q\big) + (B - 1 - c),$$

and because $0 \le B - 1 - c < B$, this exhibits $B^t - 1 - x$ in the same $(Bq' + c')$ form with $q' = B^{t-1} - 1 - q$ and $c' = B - 1 - c$. By Lemma 3.1,

$$g_t(B^t - 1 - x) = (B - 1)\big|q' - R_{t-1}c'\big| = (B - 1)\big|(B^{t-1} - 1 - q) - R_{t-1}(B - 1 - c)\big|.$$

Now $R_{t-1}(B-1) = B^{t-1} - 1$, so the bracket simplifies to $(B^{t-1} - 1 - q) - (B^{t-1} - 1) + R_{t-1}c = R_{t-1}c - q$, whose absolute value is $|q - R_{t-1}c|$. Hence $g_t(B^t - 1 - x) = (B - 1)|q - R_{t-1}c| = g_t(x)$. □

Write $R = R_t$ and $\lambda = R_{t-1} + 1$; note that since $R_t = BR_{t-1} + 1$, we have $R_{t-1} \equiv -B^{-1} \pmod R$ wherever this makes sense, and more directly $\lambda \equiv 1 - B^{t-1} \pmod R$ (this will not be needed explicitly, but explains where λ comes from). For $v \in \mathbb{Z}/R\mathbb{Z}$, let $\kappa_R(v)$ denote the canonical representative of the class $\{v, -v\}$, namely $\min(v, R - v)$.

Theorem 3.3 (Semiconjugacy)**.** *For every $m \ge 0$,*

$$\kappa_R\left(\frac{g_t\big((B-1)m\big)}{B-1}\right) \;=\; \kappa_R(\lambda m) \pmod R.$$

Consequently, the projection $x \mapsto \kappa_R(x/(B-1))$ carries the local dynamics of g_t to the dynamics of multiplication by λ on $\mathbb{Z}/R\mathbb{Z}$ folded by the identification $v \sim -v$, and this correspondence respects periodic-orbit structure: periodic points map to periodic points, and the minimal period of the image class divides the minimal period of the original periodic point.

Proof. Since $R_t = BR_{t-1} + 1$, we have $R_t \equiv 1 \pmod B$, so $\gcd(R_t, B) = 1$; this coprimality is used at the end of the argument to cancel a factor of B.

Write $y = (B-1)m$, and let $q = \lfloor y/B \rfloor$, $c = y \bmod B$, so $y = Bq + c$ and, by Lemma 3.1, $g_t(y) = (B-1)v$ where $v = |q - R_{t-1}c|$; this already confirms $g_t(y)/(B-1) = v$ is an integer, so the left side of the claimed identity is well defined.

We record two arithmetic identities that follow from the definitions of $y, \lambda, R = R_t$:

$$Bq = By - c = (B-1)Bm - c = Bm - m - c, \tag{1}$$

where the last equality just uses $Bq + c = y = (B-1)m$, i.e. $Bq = (B-1)m - c = Bm - m - c$; and, using $R = R_t = BR_{t-1} + 1$ and $\lambda = R_{t-1} + 1$,

$$B\lambda = BR_{t-1} + B = (R - 1) + B = R + (B - 1), \tag{2}$$

so that $B\lambda m = Rm + (B-1)m = Rm + Bm - m$.

Now split into the two cases occurring in the absolute value defining v.

Case $q \ge R_{t-1}c$. Here $v = q - R_{t-1}c$. Multiplying by B and substituting (1),

$$Bv = Bq - BR_{t-1}c = (Bm - m - c) - c\,(R - 1) = Bm - m - cR,$$

using $BR_{t-1} = R - 1$. On the other hand $B\lambda m = Rm + Bm - m$ from (2). Subtracting,

$$B\lambda m - Bv = \big(Rm + Bm - m\big) - \big(Bm - m - cR\big) = Rm + cR = R(m + c),$$

so $B\lambda m \equiv Bv \pmod R$. Since $\gcd(B, R) = 1$ we may cancel B, giving $\lambda m \equiv v \pmod R$, hence $\kappa_R(v) = \kappa_R(\lambda m)$.

Case $q < R_{t-1}c$. Here $v = R_{t-1}c - q$, so

$$Bv = BR_{t-1}c - Bq = c(R-1) - (Bm - m - c) = cR - Bm + m,$$

again using $BR_{t-1} = R - 1$ and (1). Adding $B\lambda m = Rm + Bm - m$ from (2),

$$Bv + B\lambda m = cR - Bm + m + Rm + Bm - m = R(m + c),$$

so $B(v + \lambda m) \equiv 0 \pmod R$, and cancelling B gives $v + \lambda m \equiv 0 \pmod R$, i.e. $v \equiv -\lambda m \pmod R$. Since κ_R identifies w with $-w$, this again gives $\kappa_R(v) = \kappa_R(\lambda m)$.

In both cases $\kappa_R(v) = \kappa_R(\lambda m)$, which is the claimed identity (recall $v = g_t(y)/(B-1)$).

For the "consequently" clause, write $\mathrm{proj}(x) = \kappa_R(x/(B-1))$, defined whenever $B - 1 \mid x$, which holds for every x in the image of g_t (Lemma 3.1) and hence for every periodic point of g_t. The identity just proved says exactly that $\mathrm{proj}(g_t(x)) = \kappa_R(\lambda(x/(B-1)))$ for x a multiple of $B - 1$. Thus proj is a semiconjugacy from the eventual dynamics of g_t to the folded multiplicative map. In particular, periodic points map to periodic points, and a factor map can reduce periods: the period of the image class divides the period of the original periodic point. Equality need not hold in general, because distinct periodic states can be identified by the folding. □

Remark. This reduction is exact, with no approximation, and two subtleties deserve emphasis. First, λ need not be a unit modulo R — for instance $B = 10, t = 3$ gives $R = 111$, $\lambda = 12$, $\gcd(\lambda, R) = 3$ — so multiplication by λ is not a bijection of $\mathbb{Z}/R\mathbb{Z}$ in general, and the periodic-orbit structure must be computed by genuine forward iteration rather than assumed to be a group action. Second, an orbit of the multiplicative dynamics may be individually symmetric under $v \mapsto -v$ (a "self-paired" orbit), in which case it folds to exactly half its upstairs length rather than folding by pairing with a distinct orbit; both phenomena occur, and both are accounted for by the argument above, since it never assumed invertibility or the absence of self-pairing. §6 works directly with the equivalent formulation $g_t(v) \equiv \pm cv \pmod{B^t - 1}$ for $c = (B-1)B^{t-1}$ acting on $\pm$-classes modulo $N = B^t - 1 = (B-1)R_t$, which is the same linearization phrased on the full modulus N rather than on the repunit $R_t = N/(B-1)$; the two are related by the bijection $v \leftrightarrow v/(B-1)$ used in the proof above.

3.1. Exact Fixed-Point Count

Theorem 3.4. *For $B \geq 2$ and $t \geq 2$, the number of fixed points of g_t on $\{0, \ldots, B^t - 1\}$ is*

$$1 + \left\lfloor \frac{(B-1) \cdot \gcd(B^{t-1} - 2,\ 2B - 1)}{2B - 1} \right\rfloor.$$

Proof. Write $A = B^{t-1} - 2 \geq 0$ (using $t \geq 2$) and $M = 2B - 1$. By definition, $g_t(x) = x$ means $\mathrm{dist}(x, \rho_t(x)) = x$, i.e. (since $\rho_t(x) \geq 0$) either $\rho_t(x) = 0$ and $x = 0$, or $\rho_t(x) = 2x$; the first case is subsumed by the second since $\rho_t(0) = 0 = 2 \cdot 0$. So $g_t(x) = x$ if and only if $\rho_t(x) = 2x$. Writing $x = Bq + c$ as before, $\rho_t(x) = q + cB^{t-1}$, so this becomes

$$q + cB^{t-1} = 2(Bq + c) \iff c(B^{t-1} - 2) = q(2B - 1) \iff cA = qM. \qquad (*)$$

So fixed points of g_t correspond exactly to pairs (q, c) with $0 \leq c < B$, $0 \leq q < B^{t-1}$, satisfying $cA = qM$; and since $x \mapsto (q, c)$ is a bijection between $\{0, \ldots, B^t - 1\}$ and such pairs, it suffices to count solutions of $(*)$.

Let $d = \gcd(A, M)$ and write $A = dA'$, $M = dM'$ with $\gcd(A', M') = 1$. Then $M \mid cA$ iff $dM' \mid cdA'$ iff $M' \mid cA'$ iff $M' \mid c$ (using $\gcd(A', M') = 1$). So $(*)$ has a solution q for a given c exactly when $M' \mid c$, and then $q = cA/M$ is the unique such solution. Thus the fixed points are in bijection with the multiples of $M' = M/d$ in $\{0, \ldots, B-1\}$, of which there are $\lfloor (B-1)/M' \rfloor + 1$ (namely $0, M', 2M', \ldots$).

It remains to check that every such c produces a valid $q < B^{t-1} = A + 2$, so that the corresponding $x = Bq + c$ really is less than B^t. Since $c \le B - 1$,

$$q = \frac{cA}{M} \le \frac{(B-1)A}{M}.$$

We must show $(B-1)A/M < A+2$, i.e. $(B-1)A < M(A+2) = MA+2M$, i.e. $(B-1-M)A < 2M$. But $M = 2B - 1$ gives $B - 1 - M = -B < 0$, so the left side is $\le 0 < 2M$ (as $A, B \ge 0$), and the inequality holds unconditionally. Hence every admissible c gives a genuine fixed point, and the count is exactly the number of multiples of $M' = M/d$ in $\{0, \dots, B-1\}$, namely

$$\left\lfloor \frac{B-1}{M'} \right\rfloor + 1.$$

Finally, $M = dM'$ gives $(B-1)/M' = (B-1)d/M$, hence

$$\left\lfloor \frac{B-1}{M'} \right\rfloor = \left\lfloor \frac{(B-1)d}{M} \right\rfloor.$$

Substituting $d = \gcd(A, M)$ gives the stated formula. □

In particular, 0 is always a fixed point.

4. The Two-Cycle Case and the Odd Part of $B + 1$

For $t = 2$, the local dynamics reduces to the spread map β.

Lemma 4.1. *For a 2-cycle block whose two digits differ by $m \in \{0, \dots, B-1\}$, one step of the local map produces a new spread*

$$\beta(m) \;=\; \big|\, 2m - (B+1) \,\big|, \qquad \beta(0) = 0.$$

Proof. Let the two digits of the block be q (more significant) and c (less significant), with current spread $m = |q - c|$. The local number is $x = Bq + c$, its rotation swaps the two digits, $\rho_t(x) = Bc + q$, so

$$g_2(x) = |x - \rho_t(x)| = |(B-1)q - (B-1)c| = (B-1)m.$$

This is a two-digit base-B number; write $(B-1)m = B\,r_1 + r_2$ with $0 \le r_2 < B$ for its digits (r_1, r_2), which are exactly the digits scattered back to the block, i.e. the new value of the pair. For $m \ge 1$ we have $(B-1)m = (m-1)B + (B-m)$ (check: $(m-1)B + (B-m) = mB - B + B - m = (B-1)m$, and $0 \le B - m < B$ since $1 \le m \le B-1$), so $r_1 = m - 1$, $r_2 = B - m$. The new spread is therefore

$$\beta(m) = |r_1 - r_2| = |(m-1) - (B-m)| = |2m - (B+1)|$$

for $m \ge 1$, and $\beta(0) = 0$ trivially, since the all-zero block is a fixed point of g_2. □

Write $B + 1 = 2^s \cdot u_0$ with u_0 odd.

Theorem 4.2. *The eventual image of β on $\{0, \dots, B-1\}$ — the set of spreads that recur under arbitrarily many iterations — is*

$$E(B) \;=\; \{0\} \cup \{\, 2^s u : u \text{ odd},\ 1 \le u \le u_0 - 2 \,\},$$

and β restricts to a genuine bijection of $E(B)$. Every starting spread reaches $E(B)$ within $s+1$ steps, and this bound is sharp. Consequently the eventual rank of β on $\{0, \dots, B-1\}$ equals

$$f(B+1) \;:=\; \frac{u_0 + 1}{2}.$$

Proof. Throughout, work modulo $N := B+1 = 2^s u_0$. For $m \geq 1$, $\beta(m) = |2m - N|$, so $\beta(m) \equiv \pm 2m \pmod{N}$; consequently $\beta(m)$ is congruent to $2m$ or $-2m$ modulo N, and since $\beta(m) \in \{0, \ldots, B-1\} = \{0, \ldots, N-2\}$ is the unique representative of its class modulo N in that range other than $N-1$ itself (which is never attained, as $2m = N-1$ or $2m = 1$ would force N or 1 odd in the wrong parity — concretely $\beta(m) = N-1$ would need $2m \in \{1, 2N-1\}$, both odd, impossible for even $2m$), β really does map $\{0, \ldots, B-1\}$ into itself.

Reaching a state of the form $2^s \cdot (\text{odd})$. We claim $2^{\min(j,s)} \mid \beta^{[j]}(m)$ for every $j \geq 0$ and every m, where $\beta^{[j]}$ denotes the j-th iterate. This is immediate by induction: $\beta^{[0]}(m) = m$ trivially has $2^0 = 1 \mid m$; and if $2^i \mid \beta^{[j]}(m)$ with $i \leq s$, then since $\beta(y) = |2y - N|$ and $2y$ and $N = 2^s u_0$ are both divisible by $2^{\min(i+1,s)}$ when $2^i \mid y$ (as $2y$ then has $2^{i+1} \mid 2y$, and N has $2^s \mid N$), their difference is divisible by $2^{\min(i+1,s)}$ too. Hence after s steps, $\beta^{[s]}(m) = 2^s w$ for some integer $w \geq 0$; one further step gives $\beta^{[s+1]}(m) = |2 \cdot 2^s w - 2^s u_0| = 2^s |2w - u_0|$, and since u_0 is odd, $|2w - u_0|$ is odd. So $\beta^{[s+1]}(m) = 2^s u$ with u odd, for every starting m: every state reaches the form $2^s u$ (odd u) within $s+1$ steps.

β permutes such states. On states of the form $2^s u$ with u odd and $1 \leq u \leq u_0 - 2$ (equivalently, since u, u_0 odd, $u < u_0$), we compute $\beta(2^s u) = |2^{s+1} u - 2^s u_0| = 2^s |2u - u_0|$. Since u_0 is odd, $|2u - u_0|$ is again odd, and since $u < u_0$ we have $2u - u_0 < u_0$, while also $2u - u_0 > -u_0$ trivially and one checks $|2u - u_0| \leq u_0 - 2$: indeed $|2u - u_0|$ and u_0 have the same parity considerations forcing $|2u - u_0| \neq u_0 - 1$ (both odd, so their possible common value $u_0 - 1$ is even — excluded) and $|2u - u_0| \neq u_0$ (would force $u = 0$ or $u = u_0$, both excluded by $1 \leq u \leq u_0 - 2$). So β maps $E(B) \setminus \{0\}$ into itself, and $\beta(0) = 0$, so β maps $E(B)$ into itself. Injectivity on the odd u in $[1, u_0 - 2]$: if $|2u - u_0| = |2u' - u_0|$ then either $u = u'$, or $2u - u_0 = -(2u' - u_0)$, i.e. $u + u' = u_0$ — impossible since u, u' are both odd and u_0 is odd (sum of two odds is even). So $u \mapsto |2u - u_0|$ is injective on odd $u \in [1, u_0 - 2]$, hence β is injective on $E(B)$; being an injective self-map of the finite set $E(B)$, it is a bijection of $E(B)$.

Sharpness and eventual image. Combining the two paragraphs above: every state reaches $E(B)$ in at most $s+1$ steps (shown above), and once in $E(B)$ the orbit stays in $E(B)$ forever, cycling under the bijection $\beta|_{E(B)}$ (shown above). Hence $E(B)$ is exactly the eventual image: it is invariant, reached from everywhere, and every one of its elements recurs (lying on a genuine cycle of the finite bijection $\beta|_{E(B)}$). Sharpness of the bound $s+1$ is witnessed already for $B = 5$ ($s = 1$, $u_0 = 3$): the state $m = 1$ satisfies $\beta(1) = |2 - 6| = 4 \notin E(5) = \{0, 2\}$, so one step is not enough, while $\beta(4) = |8 - 6| = 2 \in E(5)$, confirming $2 = s + 1$ steps are needed and no fewer.

The rank. By definition $|E(B)| = 1 + |\{u \text{ odd} : 1 \leq u \leq u_0 - 2\}|$. The odd integers in $[1, u_0 - 2]$ are $1, 3, \ldots, u_0 - 2$ (recall u_0 is odd, so $u_0 - 2$ is odd, $u_0 - 1$ is even and excluded), and there are $(u_0 - 1)/2$ of them. So $|E(B)| = 1 + (u_0 - 1)/2 = (u_0 + 1)/2 = f(B+1)$. □

In particular β collapses every spread to 0 if and only if $u_0 = 1$, i.e. $B+1$ is a power of two.

Theorem 4.3 (Identification with A003602)**.** *For every $B \geq 2$, the eventual rank of the two-cycle local map β equals $a(B+1)$, where a is Kimberling's paraphrase of the binary number system, OEIS sequence A003602 [7, 5]. Equivalently, writing $B + 1 = 2^s(2k-1)$, the eventual rank is k.*

Proof. Sequence A003602 is given by the closed form $a(n) = (\text{odd}(n) + 1)/2$, where $\text{odd}(n)$ is the largest odd divisor of n; equivalently $a(n) = k$ where $n = 2^j(2k-1)$ is the unique such factorization. This is identically the formula $f(n) = (u_0 + 1)/2$ of Theorem 4.2, evaluated at $n = B + 1$, since $u_0 = \text{odd}(B+1)$ by definition. □

The identification with A003602 is immediate from the formula: $a(n) = (\text{odd}(n) + 1)/2$. The dynamical content is the 2-adic decomposition in Theorem 4.2.

5. Total Collapse

We determine when the local map g_t is eventually zero.

Theorem 5.1. *Fix $B \geq 2$ and $t \geq 1$. Every state of a length-t cycle's local dynamics eventually reaches 0 if and only if $t = 1$, or $t = 2$ and $B + 1$ is a power of two.*

The case $t = 1$ is immediate, since $g_1 \equiv 0$ (a 1-cycle's local number equals its own rotation, so the subtraction always vanishes). The case $t = 2$ is exactly the content of Theorem 4.2: collapse happens if and only if $E(B) = \{0\}$, i.e. $u_0 = 1$, i.e. $B + 1$ is a power of two. The substance of Theorem 5.1 is therefore the converse for $t \geq 3$ (and the "only if" direction for $t = 2$, already covered): for every $t \geq 3$, and for $t = 2$ whenever $B + 1$ is not a power of two, some local value never reaches 0.

Proof of the remaining direction. The idea is to find, for each offending t, a divisor $d > 1$ of $M_t = B^t - 1$ coprime to $B - 1$; such a d then obstructs collapse, as we show first.

The obstruction. Right cyclic rotation of a t-digit string is exactly multiplication by B^{-1} modulo $M_t = B^t - 1$, since $B \cdot \rho_t(x) = B(q + cB^{t-1}) = Bq + cB^t \equiv Bq + c = x \pmod{M_t}$ (using $B^t \equiv 1$). Hence $B \cdot (x - \rho_t(x)) = Bx - B\,\rho_t(x) \equiv Bx - x = (B-1)x \pmod{M_t}$, so for the local map $g_t(x) = |x - \rho_t(x)|$ we get $B\,g_t(x) \equiv \pm(B-1)x \pmod{M_t}$ (the sign depending on which of $x, \rho_t(x)$ is larger). Iterating j times and tracking signs,

$$B^j\, g_t^j(x) \equiv \pm(B-1)^j\, x \pmod{M_t}.$$

Now let d be any divisor of M_t with $\gcd(d, B-1) = 1$. Since $d \mid M_t = B^t - 1$, also $\gcd(d, B) = 1$ (a common factor of d and B would divide B^t and $M_t = B^t - 1$, forcing it to divide 1). So both B and $B - 1$ are invertible modulo d, and reducing the displayed congruence modulo d (which divides M_t) and cancelling the unit B^j shows

$$g_t^j(x) \equiv \pm(B-1)^j (B^{-1})^j\, x \pmod{d},$$

which is x times a unit modulo d; in particular $d \mid g_t^j(x)$ if and only if $d \mid x$. So divisibility by d is a conserved quantity of the dynamics: if $d \nmid x$ for the starting local value x, then $d \nmid g_t^j(x)$ for every $j \geq 0$, and since d divides every multiple of d including 0, in particular $g_t^j(x) \neq 0$ for every j: this local value never reaches 0. So the existence of such a $d > 1$ obstructs total collapse.

Constructing d when t has an odd prime factor. Suppose $t = nq$ with q an odd prime and $n \geq 1$. Set $C = B^n \geq 2$ and $M = 1 + C + \cdots + C^{q-1}$, so that $M(C-1) = C^q - 1 = B^t - 1 = M_t$, i.e. $M \mid M_t$.

Since $C \equiv 1 \pmod{C-1}$, each power $C^i \equiv 1 \pmod{C-1}$, so $M \equiv q \pmod{C-1}$. Let $e = \gcd(M, B-1)$. Because $B - 1 \mid C - 1$ (as $C - 1 = B^n - 1$ is a multiple of $B - 1$) and $e \mid B - 1$, we get $e \mid C - 1$; reducing $M \equiv q \pmod{C-1}$ modulo the further divisor e gives $M \equiv q \pmod{e}$, and since $e \mid M$ this forces $e \mid q$. As q is prime, $e \in \{1, q\}$.

If $e = 1$, take $d = M$. Since $M = 1 + C + \cdots + C^{q-1} \geq 2^q - 1$ (each of the q terms is at least the corresponding term of $1 + 2 + \cdots + 2^{q-1}$, as $C \geq 2$), and $2^q - 1 > 1$ for $q \geq 2$, we have $d = M > 1$ and $\gcd(d, B-1) = e = 1$, as required.

If $e = q$, then in particular $q \mid B - 1$, hence $q \mid C - 1$ (again using $B - 1 \mid C - 1$); this also gives $q \nmid C$ (else $q \mid C - (C-1) = 1$). Since q is odd and $q \mid C - 1$ with $q \nmid C$, the Lifting-the-Exponent lemma applies to $C^q - 1^q$ at the prime q:

$$v_q(C^q - 1) = v_q(C - 1) + v_q(q) = v_q(C - 1) + 1.$$

On the other hand $C^q - 1 = M(C-1)$, so $v_q(M) + v_q(C-1) = v_q(C-1) + 1$, giving $v_q(M) = 1$ exactly. Thus M/q is an integer coprime to q, and since every common divisor of M and $B - 1$ divides q (shown above) while M contains only a single factor of q, M/q is coprime to $B - 1$. Take $d = M/q$; as before $M \geq 2^q - 1 > q$ for every prime $q \geq 2$, so $d = M/q > 1$.

In either case we obtain $d > 1$ dividing M_t with $\gcd(d, B-1) = 1$.

Constructing d when t is a power of two. Write $t = 2n$ with $n \geq 2$ (this covers all remaining $t \geq 3$, i.e. $t = 4, 8, 16, \ldots$, once the odd-prime-factor case above has covered every t with an odd prime factor — together these two cases exhaust all $t \geq 3$, since an integer ≥ 3 either has an odd prime factor or is a power of two with exponent ≥ 2). Then

$$M_t = B^t - 1 = (B^n - 1)(B^n + 1).$$

Since $B-1 \mid B^n - 1$, any common divisor of B^n+1 and $B-1$ must divide their difference-type combination; concretely, $\gcd(B^n+1, B-1)$ divides $\gcd(B^n+1, B^n-1)$, and $\gcd(B^n+1, B^n-1) \mid 2$ (their difference is 2), so $\gcd(B^n+1, B-1) \mid 2$. Let d be the odd part of B^n+1 (i.e. B^n+1 with all factors of 2 removed); then $d \mid B^n+1 \mid M_t$, and $\gcd(d, B-1)$ divides $\gcd(B^n+1, B-1)$ which divides 2, but d is odd, so $\gcd(d, B-1) = 1$. Finally $d > 1$. If B is even, then B^n+1 is odd and at least 5, so $d = B^n + 1 > 1$. If B is odd and n is even, then $B^n \equiv 1 \pmod 4$, so $B^n + 1 \equiv 2 \pmod 4$ and $d = (B^n+1)/2 > 1$. If B is odd and n is odd, then $n \geq 3$ and

$$B^n + 1 = (B+1)(B^{n-1} - B^{n-2} + \cdots - B + 1),$$

where the second factor is an odd integer greater than 1; hence the odd part of $B^n + 1$ is greater than 1.

Every $t \geq 3$ has either an odd prime factor or is a power of two with exponent ≥ 2, so one of the two constructions above always applies, producing the required $d > 1$ dividing M_t and coprime to $B-1$. By the obstruction paragraph, taking $x = 1$ (which is not divisible by any $d > 1$) shows the local value $x = 1$ never reaches 0 under iteration of g_t, for every $B \geq 2$ and $t \geq 3$. This completes the proof, together with the $t = 2$ case already covered by Theorem 4.2. □

Corollary 5.2. *If some cycle of π has length $t \geq 3$, then F_π is not eventually the constant zero map: some digit vector never reaches 0 under iteration.*

Proof. By Theorem 5.1, the local value 1 (or any nonzero value not divisible by the witnessing d) never reaches 0 under g_t. By Theorem 2.1, the digit vector that reads this local value on the offending cycle and 0 elsewhere never reaches the all-zero vector under F_π, since the offending cycle's coordinate never becomes 0. □

6. Periodic Structure of the Local Map

We now determine the periodic points, their individual periods, the common eventual period of the periodic set, and the transient in terms of the arithmetic of $B^t - 1$.

Throughout this section fix $t \geq 1$ and $B \geq 2$, and put

$$N \;=\; B^t - 1, \qquad c \;=\; (B-1)\, B^{t-1}.$$

This is the full-modulus form of the linearization in Theorem 3.3.

6.1. The Linear Model on ±-Classes

Lemma 6.1. *$g_t(N - v) = g_t(v)$ for every $0 \leq v \leq N$, and $g_t(v) \equiv \pm c\, v \pmod N$.*

Proof. The first claim is Lemma 3.2 restated on the modulus $N = B^t - 1$ (noting $B^t - 1 - x = N - x$). The second is the congruence established in the proof of Theorem 5.1: $B\, g_t(v) \equiv \pm(B-1)v \pmod N$, and since B is invertible modulo N (as $\gcd(B, N) = \gcd(B, B^t - 1) = 1$) with inverse B^{t-1} (because $B \cdot B^{t-1} = B^t \equiv 1 \pmod N$), multiplying both sides by B^{t-1} gives $g_t(v) \equiv \pm(B-1)B^{t-1}v = \pm cv \pmod N$. □

As in Theorem 3.3, let $\kappa_N(v)$ denote the canonical representative of the class $\{v, -v\}$ modulo N. Lemma 6.1 says exactly that g_t is semiconjugate, via κ_N, to the linear map $q \mapsto \kappa_N(cq)$ on $\pm$-classes modulo N:

$$\kappa_N(g_t(v)) \;=\; \kappa_N(c \cdot \kappa_N(v)).$$

Proposition 6.2. *The map κ_N restricts to a bijection between the periodic points of g_t and the periodic $\pm$-classes of multiplication by c modulo N.*

Proof. First suppose v, w are periodic points of g_t with $\kappa_N(v) = \kappa_N(w)$. Since a periodic point cannot equal N (indeed $g_t(N) = 0$), we have $v, w \in \{0, \dots, N-1\}$. Thus either $v = w$ or $w = N - v$. In the second case Lemma 6.1 gives $g_t(v) = g_t(w)$. But g_t restricts to a bijection of its finite periodic set, hence is injective there. Therefore $v = w$. So κ_N is injective on periodic points.

Conversely, let C be a periodic $\pm$-class of the multiplicative model. Choose x with $\kappa_N(x) = C$. Since the domain of g_t is finite, some forward iterate of x reaches a periodic point p. By semiconjugacy, the class $\kappa_N(p)$ lies on the same periodic orbit of the multiplicative model as C. The image of the periodic cycle containing p under κ_N is a periodic cycle of the multiplicative model; since the period of the image cycle divides the period of the original cycle, that image is the entire periodic orbit containing C. Hence some periodic point on the cycle containing p maps to C.

Thus every periodic multiplicative $\pm$-class is represented by a periodic point of g_t, proving surjectivity and hence the bijection. □

6.2. The Coprime Splitting and the Exact Count of Periodic Points

Definition 6.3. For $m, N \geq 1$, a splitting $N = N_1 N_2$ is an *m-split* if every prime factor of N_1 divides m and $\gcd(N_2, m) = 1$.

Such a factorization exists and is unique for every $N \geq 1$ and $m \geq 1$: repeatedly divide out from N any prime factor shared with m, which terminates since N is finite, producing N_1 (the product of primes removed, with multiplicity) and N_2 (what remains, coprime to m by construction); uniqueness follows because any divisor of N coprime to m must divide the coprime part of any valid splitting, so two splittings have the same N_2, hence the same $N_1 = N/N_2$. We apply this with $m = B - 1$ and $N = B^t - 1$, and write

$$N \;=\; N_1 \cdot N_2, \qquad \text{every prime of } N_1 \text{ divides } B-1, \qquad \gcd(N_2, B-1) = 1.$$

Remark (The degenerate case $B = 2$). When $B = 2$ we have $m = B - 1 = 1$, so every $N \geq 1$ is trivially 1-split as $N_1 = 1$, $N_2 = N$: the splitting carries no information, and correspondingly $a_0 = 0$ in Theorem 6.11 for every t (the least $a \geq 0$ with $1 \mid c^a$ is $a = 0$), so the transient collapses immediately and every point is periodic after at most one step. The theory below is not vacuous in this case, only degenerate in the expected way: $|\operatorname{Per}(g_t)| = (N+1)/2$ with $N = 2^t - 1$, consistent with the binary case worked out by Yamagami [10].

Lemma 6.4. *N_2 is odd.*

Proof. If B is odd then $B - 1$ is even, and every prime of N_1 divides $B - 1$, so if $2 \mid N_2$ then $2 \mid \gcd(N_2, B-1) = 1$, a contradiction; hence N_2 is odd. If B is even then $B - 1$ is odd, and $N = B^t - 1$ is odd (as B^t is even), so $N_2 \mid N$ is automatically odd. □

Theorem 6.5. *A $\pm$-class q modulo N is periodic under multiplication by c if and only if $N_1 \mid q$.*

Proof. Write $N = N_1 N_2$ with $\gcd(N_1, N_2) = 1$: every prime factor of N_1 divides $B-1$ (to its full multiplicity in N), and N_2 is the complementary, $(B-1)$-coprime factor. Two preliminary facts:

(a) $\gcd(c, N_2) = 1$. A common prime factor of $c = (B-1)B^{t-1}$ and N_2 would divide $B-1$ or B. But $\gcd(N_2, B-1) = 1$ by construction, and $\gcd(N_2, B) = 1$ since $N_2 \mid N = B^t - 1$ (a prime dividing both B and $B^t - 1$ would divide 1). So no such common prime exists.

(b) $N_1 \mid c^n$ *for every* $n \geq N_1$. Every prime $p \mid N_1$ divides $B-1$, hence divides $c = (B-1)B^{t-1}$. Comparing p-adic valuations, once the exponent n is at least N_1 it certainly exceeds $v_p(N_1)$ for every prime $p \mid N_1$ (as $v_p(N_1) \leq \log_2 N_1 < N_1$), so $v_p(c^n) = n \cdot v_p(c) \geq n \geq N_1 > v_p(N_1)$ for every such p, while for primes $p \nmid N_1$ the valuation condition is vacuous. Hence $N_1 \mid c^n$.

If $N_1 \mid q$. Write $q = N_1 r$. By fact (a), c is invertible modulo N_2, hence (as $\mathbb{Z}/N_2\mathbb{Z}$ is finite) has finite multiplicative order there; by Euler's theorem, $c^{\varphi(N_2)} \equiv 1 \pmod{N_2}$, where φ is Euler's totient function. We claim $c^{\varphi(N_2)} q \equiv q \pmod{N}$. Since $q = N_1 r$, this congruence modulo $N = N_1 N_2$ splits (Chinese Remainder Theorem) into a congruence modulo N_1 and one modulo N_2. Modulo N_1: both $c^{\varphi(N_2)} q$ and q are multiples of N_1, hence congruent to 0. Modulo N_2: $c^{\varphi(N_2)} q \equiv 1 \cdot q = q$ by Euler's theorem. Both congruences hold, so $c^{\varphi(N_2)} q \equiv q \pmod{N}$, and q is periodic (with the + sign, period dividing $\varphi(N_2)$).

If q *is periodic.* Then $c^j q \equiv \varepsilon q \pmod{N}$ for some $j \geq 1$ and $\varepsilon \in \{+1, -1\}$, i.e. $N \mid c^j q - \varepsilon q$ (as integers, allowing $c^j q - \varepsilon q$ to be negative when $\varepsilon = 1$ and $q > c^j q$). Squaring removes the sign: from the integer identity

$$(c^j)^2 q - q \;=\; c^j\,(c^j q - \varepsilon q) + \varepsilon\,(c^j q - \varepsilon q) + (\varepsilon^2 - 1)q,$$

and $\varepsilon^2 = 1$, the last term vanishes, so $(c^j)^2 q - q = (c^j + \varepsilon)(c^j q - \varepsilon q)$, which is divisible by N since $c^j q - \varepsilon q$ is. Hence

$$c^{2j} q \equiv q \pmod{N}.$$

Iterating this relation m times (each step multiplies both sides by c^{2j}, using $c^{2j} \cdot (c^{2j})^m q \equiv c^{2j} q \equiv q$, so by induction $(c^{2j})^m q \equiv q \pmod{N}$ for every $m \geq 0$) gives $c^{2jm} q \equiv q \pmod{N}$ for every $m \geq 0$.

By fact (b), choosing m large enough that $2jm \geq N_1$, we have $N_1 \mid c^{2jm}$, so $c^{2jm} q \equiv 0 \pmod{N_1}$. Combined with $c^{2jm} q \equiv q \pmod{N}$ (hence also modulo N_1, since $N_1 \mid N$), this gives $q \equiv 0 \pmod{N_1}$, i.e. $N_1 \mid q$, as required. □

Theorem 6.6 (Count of periodic points)**.** *The local map* g_t *has exactly*

$$|\operatorname{Per}(g_t)| \;=\; \frac{N_2 + 1}{2}$$

periodic points on $\{0, \ldots, B^t - 1\}$*, where* N_2 *is the* $(B-1)$*-coprime part of* $N = B^t - 1$*; this quantity is an integer since* N_2 *is odd (Lemma 6.4).*

Proof. By Proposition 6.2, the periodic points of g_t are in bijection with the periodic $\pm$-classes of multiplication by c modulo N, which by Theorem 6.5 are exactly the $\pm$-classes of multiples of N_1. Among the representatives in $\{0, \ldots, N-1\}$ these are

$$0, N_1, 2N_1, \ldots, (N_2 - 1)N_1,$$

so there are N_2 such representatives. Since N_2 is odd, 0 is the unique class fixed by $q \mapsto -q$ among them: if $N_1 r \equiv -N_1 r \pmod{N}$, then $2r \equiv 0 \pmod{N_2}$, forcing $r \equiv 0 \pmod{N_2}$. Thus 0 contributes one class and the remaining $N_2 - 1$ representatives pair up, giving

$$1 + \frac{N_2 - 1}{2} = \frac{N_2 + 1}{2}$$

periodic $\pm$-classes. The same number of periodic points follows from Proposition 6.2. □

Remark (Consistency with the direct two-cycle computation). For $t = 2$, Theorem 6.6 gives $|\operatorname{Per}(g_2)| = (N_2 + 1)/2$ where $B^2 - 1 = N_1 N_2$ is the $(B-1)$-coprime splitting, while the independent, purely combinatorial argument of §4 (Theorem 4.2) gives the eventual rank of the spread map β as $f(B+1) = (u_0+1)/2$ with $B+1 = 2^s u_0$. These two formulas are derived by entirely different routes — one via the arithmetic of $\pm$-classes modulo $B^t - 1$, the other via direct 2-adic tracking of the spread — and agree on every case checked (e.g. $B = 10$: $N_2 = 11$ gives 6, and $u_0 = 11$ gives $f(11) = 6$; the values of N_2 for $B = 3, 4, 5, 10$ recorded in Table 2 likewise match $\operatorname{odd}(B+1)$ in each case). This agreement is not a coincidence — both count the eventual image of the same map g_2 under different presentations — but it is not immediate from the statements alone, since N_2 and $u_0 = \operatorname{odd}(B+1)$ are a priori different numbers; the reader can check $N_2 = u_0$ whenever the $(B-1)$-coprime part of $B^2 - 1 = (B-1)(B+1)$ is computed out, which follows from $\gcd(B-1, B+1) \mid 2$ and N_2 odd (Lemma 6.4).

Theorem 6.7 (Eventual image)**.** *For every $j \geq B^t$, the image of $\{0, \ldots, B^t - 1\}$ under g_t^j equals $\operatorname{Per}(g_t)$, the set of periodic points. Consequently the eventual rank of g_t is $(N_2+1)/2$.*

Proof. The domain has B^t elements, so by pigeonhole every orbit enters a cycle within B^t steps and stays there; hence for $j \geq B^t$ the image of g_t^j is contained in, and by periodicity equal to, the set of points lying on cycles, which is exactly $\operatorname{Per}(g_t)$. This bound is not sharp and is improved to a much smaller two-value transient bound in Theorem 6.11 below. □

6.3. The Period: An Explicit Multiplicative Order

Theorem 6.8 (Individual periods)**.** *Let $q = N_1 r$ represent a periodic $\pm$-class modulo N, with $0 \leq r < N_2$. If $r = 0$, the corresponding periodic point is the fixed point 0. For $r \neq 0$, put*

$$d_r := \gcd(r, N_2), \qquad M_r := \frac{N_2}{d_r}.$$

Then the minimal period of the corresponding periodic $\pm$-class is

$$P(r) = \min\{j \geq 1 : c^j \equiv \pm 1 \pmod{M_r}\}.$$

In particular, if

$$P_* := \min\{j \geq 1 : c^j \equiv \pm 1 \pmod{N_2}\},$$

then P_* is a period of the entire periodic set and is the least common multiple of all individual periodic-orbit lengths. Equivalently,

$$P_* = \operatorname{ord}_{(\mathbb{Z}/N_2\mathbb{Z})^\times/\{\pm 1\}}(c).$$

Proof. Let v be periodic and write $q = \kappa_N(v) = N_1 r$. The return condition for the folded multiplicative model is

$$c^j q \equiv \pm q \pmod{N}.$$

Modulo N_1 this is automatic because q is a multiple of N_1. Modulo N_2, since N_1 is invertible, it is equivalent to

$$c^j r \equiv \pm r \pmod{N_2}.$$

Write $r = d_r u$ with $\gcd(u, N_2/d_r) = 1$. Dividing by d_r gives

$$c^j u \equiv \pm u \pmod{N_2/d_r},$$

and now u is invertible modulo N_2/d_r, so this is equivalent to

$$c^j \equiv \pm 1 \pmod{N_2/d_r}.$$

Therefore the minimal period is exactly the stated $P(r)$.

For P_*, take $r = 1$. Then

$$P_* = \min\{j \geq 1 : c^j \equiv \pm 1 \pmod{N_2}\}.$$

Because every modulus N_2/d_r divides N_2, the congruence $c^{P_*} \equiv \pm 1 \pmod{N_2}$ implies the corresponding congruence modulo every N_2/d_r. Hence P_* is a period of every periodic orbit. Conversely, any common period of all periodic orbits must in particular be a period of the class represented by N_1, for which $r = 1$, so it must be a multiple of P_*. Thus P_* is the least common multiple of all individual periods. The final characterization is the definition of the order of the coset of c in $(\mathbb{Z}/N_2\mathbb{Z})^\times/\{\pm 1\}$. □

Example 6.9. For $B = 7$, $t = 3$: $N = B^t - 1 = 342 = 2 \cdot 3^2 \cdot 19$, and since $B - 1 = 6 = 2 \cdot 3$, the $(B-1)$-coprime splitting is $N_1 = 18 = 2 \cdot 3^2$, $N_2 = 19$. By Theorem 6.6, the local map has $(19 + 1)/2 = 10$ periodic points. The constant is $c = (B-1)B^{t-1} = 6 \cdot 49 = 294 \equiv 294 - 15 \cdot 19 = 294 - 285 = 9 \pmod{19}$. We compute the successive powers of 9 modulo 19 directly:

$$9^1 \equiv 9, \quad 9^2 \equiv 5, \quad 9^3 \equiv 7, \quad 9^4 \equiv 6, \quad 9^5 \equiv 16 \equiv -3,$$
$$9^6 \equiv 11 \equiv -8, \quad 9^7 \equiv 4, \quad 9^8 \equiv 17 \equiv -2, \quad 9^9 \equiv 1 \pmod{19}.$$

None of $9^1, \ldots, 9^8$ is congruent to ± 1 modulo 19, while $9^9 \equiv 1$. Hence

$$P_* = 9$$

is the order of ± 9 in $(\mathbb{Z}/19\mathbb{Z})^\times/\{\pm 1\}$. Since $N_2 = 19$ is prime, every nonzero periodic class has $r \not\equiv 0 \pmod{19}$ and hence has minimal period 9. Thus the periodic set consists of the fixed point 0 together with one 9-cycle.

Example 6.10 (Multiple periodic lengths)**.** The individual periods need not all be equal. Take $B = 6$ and $t = 4$. Then

$$N = 6^4 - 1 = 1295 = 5 \cdot 259, \qquad N_1 = 5, \qquad N_2 = 259,$$

and there are 130 periodic points. Here $c = 5 \cdot 6^3 = 1080 \equiv 44 \pmod{259}$ and

$$P_* = \min\{j \geq 1 : 44^j \equiv \pm 1 \pmod{259}\} = 9.$$

However, different periodic classes have different minimal periods. For example,

$$185 \longmapsto 925 \longmapsto 555 \longmapsto 185$$

is a 3-cycle, while

$$5 \longmapsto 1075 \longmapsto 680 \longmapsto 135 \longmapsto 535 \longmapsto 230 \longmapsto 240 \longmapsto 200 \longmapsto 265 \longmapsto 5$$

is a 9-cycle. The fixed point 0 gives period 1. Thus the periodic set has cycle lengths 1, 3, and 9, while $9 = P_*$ is the least common multiple of all of them. This illustrates the divisor-dependent formula in Theorem 6.8.

6.4. The Transient

Theorem 6.11 (Transient length)**.** *Let a_0 be the least $a \geq 0$ with $N_1 \mid c^a$ (this exists, since every prime factor of N_1 divides $B - 1$, hence divides $c = (B-1)B^{t-1}$, so $N_1 \mid c^a$ once a exceeds the largest exponent occurring in the prime factorization of N_1). Then:*

(i) every state is periodic after $a_0 + 1$ steps, i.e. $g_t^{[a_0+1]}(\{0, \ldots, B^t - 1\}) = \mathrm{Per}(g_t)$;

(ii) if $a_0 \geq 1$ (equivalently, $N_1 \nmid c^{a_0-1}$, since a_0 is by definition the least exponent with $N_1 \mid c^{a_0}$), the image at step $a_0 - 1$ is not yet the full attractor.

Consequently the transient length ι (the least j with $g_t^j(\{0, \ldots, B^t - 1\}) = \mathrm{Per}(g_t)$) satisfies $a_0 \leq \iota \leq a_0 + 1$.

Proof. Throughout, write $N = B^t - 1 = N_1 N_2$, and recall from Theorem 6.5 that a class $q < N$ lies in $\mathrm{Per}_{\pm}(N, c)$ (is periodic for multiplication by c) iff $N_1 \mid q$.

Part (i). Fix $x \leq N$ and set $y = g_t^{[a]}(x)$; we must show $g_t(y) \in \mathrm{Per}(g_t)$. Two cases:

- If $y = N$ (the all-$(B-1)$ string), then by the complementation symmetry of Lemma 3.2 applied at $v = 0$, $g_t(N) = g_t(0\text{'s complement}) = 0$, and $0 \in \mathrm{Per}(g_t)$ directly (it is a fixed point: $g_t(0) = 0$).

- If $y < N$, the key computation is that class-taking intertwines $g_t^{[a]}$ with the multiplicative map: $\kappa_N(y) = \kappa_N(g_t^{[a]}(x)) = \kappa_N(c^a \cdot \kappa_N(x))$. Since $N_1 \mid c^a$ by hypothesis, N_1 divides $c^a \cdot \kappa_N(x)$, and because $N_1 \mid N$, divisibility by N_1 passes through reduction mod N and through the class map unchanged (this is exactly where $N_1 \mid N$ is used: $N_1 \mid \kappa_N(w) \iff N_1 \mid w$ for any w). Hence $N_1 \mid \kappa_N(y)$, i.e. $\kappa_N(y) \in \mathrm{Per}_{\pm}(N, c)$ by Theorem 6.5. By the surjectivity half of the bijection in Proposition 6.2, every periodic class is realized by an actual periodic point $v \in \mathrm{Per}(g_t)$ with $\kappa_N(v) = \kappa_N(y)$; since $y, v < N$ share a class, either $y = v$ or $y + v = N$ (the two ways two representatives below N can agree after folding). In the first case $y = v \in \mathrm{Per}(g_t)$ and $g_t(y) \in \mathrm{Per}(g_t)$ by invariance of the attractor (Theorem 6.5: g_t maps $\mathrm{Per}(g_t)$ to itself). In the second case $y = N - v$ is the complement of the periodic point v, so by Lemma 3.2, $g_t(y) = g_t(v) \in \mathrm{Per}(g_t)$ again.

Either way $g_t^{[a+1]}(x) = g_t(y) \in \mathrm{Per}(g_t)$, proving (i) (and, applying this at every $j \geq a+1$ via one further induction on $j - (a+1)$ using invariance of $\mathrm{Per}(g_t)$, the whole image $g_t^j(\{0, \ldots, B^t - 1\})$ equals $\mathrm{Per}(g_t)$ for every such j, not merely $\subseteq$, since $\mathrm{Per}(g_t) \subseteq$ every such image by definition of periodicity).

Part (ii). Suppose $a_0 \geq 1$, so that by minimality of a_0 we have $N_1 \nmid c^{a_0-1}$; set $a = a_0 - 1$. Take $x = 1$, whose class is $\kappa_N(1) = 1$ (as $1 < N/2$ for $N > 2$). As in (i), $\kappa_N(g_t^{[a]}(1)) = \kappa_N(c^a \cdot 1) = \kappa_N(c^a)$. If $g_t^{[a]}(1)$ were in $\mathrm{Per}(g_t)$, its class would lie in $\mathrm{Per}_{\pm}(N, c)$, forcing $N_1 \mid c^a = c^{a_0-1}$ by Theorem 6.5 — contradicting $N_1 \nmid c^{a_0-1}$. Hence $g_t^{[a_0-1]}(1) \notin \mathrm{Per}(g_t)$, so in particular the image $g_t^{[a_0-1]}(\{0, \ldots, B^t - 1\})$, which contains $g_t^{[a_0-1]}(1)$, is not contained in $\mathrm{Per}(g_t)$ and so cannot equal it.

Combining: a_0 (least a with $N_1 \mid c^a$) makes the image collapse to the full attractor by step $a_0 + 1$ via (i), while (ii) shows the image at step $a_0 - 1$ is not yet the attractor whenever $a_0 \geq 1$. Hence $\iota > a_0 - 1$, i.e. $\iota \geq a_0$, when $a_0 \geq 1$; when $a_0 = 0$ the bound $\iota \geq a_0$ is trivial. Together with $\iota \leq a_0 + 1$ from (i), this pins the exact transient ι to $a_0 \leq \iota \leq a_0 + 1$. □

6.5. Worked Instances

Table 1 lists several instances of Theorem 6.6, Theorem 6.8, and Theorem 6.11.

The preceding results give the following description.

Theorem 6.12 (Complete structure of the attracting cycles)**.** *Fix $B \geq 2$ and $t \geq 1$, and write $B^t - 1 = N_1 N_2$ for the $(B-1)$-coprime splitting and $c = (B-1)B^{t-1}$. Then the local map g_t has exactly*

$$\frac{N_2 + 1}{2}$$

periodic points. If a periodic point corresponds under Proposition 6.2 to the $\pm$-class of $N_1 r$, then for $r \neq 0$ its minimal period is

$$P(r) = \min\left\{ j \geq 1 : c^j \equiv \pm 1 \pmod{N_2 / \gcd(r, N_2)} \right\},$$

B	t	$B^t - 1 = N_1 \cdot N_2$	periodic points $(N_2+1)/2$	a_0	P_*
10	2	$99 = 9 \cdot 11$	6	1	5
7	3	$342 = 18 \cdot 19$	10	2	9
10	3	$999 = 27 \cdot 37$	19	2	9
13	3	$2196 = 36 \cdot 61$	31	2	15
14	3	$2743 = 13 \cdot 211$	106	1	105
6	4	$1295 = 5 \cdot 259$	130	1	9

Table 1: Worked instances of Theorem 6.6, Theorem 6.8, and Theorem 6.11, consolidated below as Theorem 6.12. The column P_* is the least common multiple of the individual periodic-orbit lengths. The $B = 10, t = 2$ row gives the 5-cycle described in §4. For $B = 14, t = 3$, $P_* = 105$. For $B = 6, t = 4$, smaller cycles coexist, as shown in the worked example following Theorem 6.8.

while 0 *is fixed. The quantity*

$$P_* := \min\{j \geq 1 : c^j \equiv \pm 1 \pmod{N_2}\}$$

is the least common multiple of all periodic-orbit lengths. Finally, the transient before every state reaches the periodic set is a_0 *or* $a_0 + 1$*, where* a_0 *is the least* a *such that* $N_1 \mid c^a$*.*

Proof. Immediate from Theorem 6.6 (count), Theorem 6.8 (period), and Theorem 6.11 (transient). □

7. Eventual Rank for an Arbitrary Permutation

The product decomposition gives the eventual rank of F_π.

Theorem 7.1 (Eventual rank of a single generator)**.** *Let* $\pi \in S_k$ *have cycles* $C_1, \ldots, C_m$ *of lengths* $t_1, \ldots, t_m$*. For every* $B \geq 2$ *and every* $j \geq \max_i B^{t_i}$*,*

$$|\operatorname{Im}(F_\pi^j)| \;=\; \prod_{i=1}^{m} \frac{N_2(B, t_i) + 1}{2},$$

where $B^{t_i} - 1 = N_1(B, t_i) \cdot N_2(B, t_i)$ *is the* $(B-1)$*-coprime splitting attached to cycle length* t_i*.*

Proof. By Theorem 2.1, coding a digit vector d by the tuple of its local values $(x_1, \ldots, x_m)$ associated with the m cycles of π is a bijection between digit vectors and tuples $\prod_i \{0, \ldots, B^{t_i} - 1\}$, and this coding conjugates F_π to the product map $g_{t_1} \times \cdots \times g_{t_m}$ acting coordinatewise. Consequently F_π^j corresponds, under this bijection, to $g_{t_1}^{[j]} \times \cdots \times g_{t_m}^{[j]}$, and the image of a product map is the product of the images of its factors:

$$\operatorname{Im}(F_\pi^j) \;\longleftrightarrow\; \prod_{i=1}^{m} \operatorname{Im}(g_{t_i}^j).$$

For $j \geq B^{t_i}$, Theorem 6.7 identifies $\operatorname{Im}(g_{t_i}^j) = \operatorname{Per}(g_{t_i})$, of size $(N_2(B, t_i) + 1)/2$ by Theorem 6.6. Taking j large enough for every cycle simultaneously (possible since there are finitely many cycles) and taking cardinalities of both sides of the bijection above gives the product formula. □

Example 7.2. For $\pi = (1\,3)(2\,4)$, both cycles have length 2, so Theorem 7.1 gives eventual rank $\left(\frac{N_2(B,2)+1}{2}\right)^2$, where $B^2 - 1 = N_1(B, 2)N_2(B, 2)$. Table 2 evaluates this for several bases.

B	$B^2-1=N_1\cdot N_2$	eventual rank of $F_{(1\,3)(2\,4)}$
3	$8=8\cdot 1$	$1^2=1$
4	$15=3\cdot 5$	$3^2=9$
5	$24=8\cdot 3$	$2^2=4$
10	$99=9\cdot 11$	$6^2=36$

Table 2: The eventual rank of $F_{(1\,3)(2\,4)}$, evaluated from Theorem 7.1.

More generally, whenever π has repeated cycle lengths — say π has e_t cycles of length t, for each t — Theorem 7.1 gives eventual rank $\prod_t \left(\frac{N_2(B,t)+1}{2}\right)^{e_t}$, a perfect power for every repeated cycle length, and in particular a perfect square whenever every cycle length of π occurs an even number of times.

Corollary 7.3 (Minimal rank and the semigroup structure)**.** *For a single full k-cycle π, the cyclic semigroup $\langle F_\pi\rangle$ contains an idempotent of rank exactly $(N_2+1)/2$ (where $B^k-1=N_1N_2$), and no element of $\langle F_\pi\rangle$ has smaller rank. Moreover, if P_* is the least common multiple of the individual periods of the periodic points of this full-cycle map, then*

$$F_\pi^{j+P_*}=F_\pi^j$$

for every sufficiently large j.

Proof. For a single full cycle, F_π is the local map g_k up to the coordinate identification. By Theorems 6.7 and 6.6, its eventual image has size exactly $(N_2+1)/2$. The ranks of the iterates form a nonincreasing sequence because

$$\mathrm{Im}(F_\pi^{j+1})\subseteq \mathrm{Im}(F_\pi^j).$$

Once the image has stabilized at the periodic set, it cannot shrink further. Hence no iterate has rank smaller than $(N_2+1)/2$.

By Theorem 6.8, P_* is a period of the entire periodic set. Choose j sufficiently large that F_π^j maps the state space onto the periodic set and with $P_*\mid j$. Then the restriction of F_π^j to the periodic set is the identity, so $(F_\pi^j)^2=F_\pi^j$. Thus an idempotent of the claimed rank exists. Since the eventual dynamics on a finite set is periodic, the same P_* gives $F_\pi^{j+P_*}=F_\pi^j$ for all sufficiently large j. □

Example 7.4. For $B=10$, $\pi=(1\,2\,3)$ (a full 3-cycle), Theorem 6.12 gives $N=999=27\cdot 37$, so the eventual rank is $(37+1)/2=19$. Here $P_*=9$, and the semigroup $\langle F_\pi\rangle$ satisfies $F_\pi^{j+9}=F_\pi^j$ for every $j\geq 3$, with every such power having rank exactly 19, matching Table 1.

8. Conclusion

For a single generator π, the cycle decomposition reduces F_π to the product of the local maps g_t. The local map is semiconjugate to multiplication by

$$c=(B-1)B^{t-1}$$

on $\pm$-classes modulo B^t-1. Writing

$$B^t-1=N_1N_2,\qquad (N_2,B-1)=1,$$

the N_1-part accounts for the transient and the N_2-part for the periodic dynamics. In particular, g_t has $(N_2+1)/2$ periodic points. Their individual periods are determined by the divisors

$N_2/\gcd(r, N_2)$ through Theorem 6.8, while the order of $\pm c$ modulo N_2 gives the least common multiple P_* of all periodic-orbit lengths. The eventual rank of g_t is $(N_2 + 1)/2$.

The collapse criterion is equally direct: every state reaches 0 only for $t = 1$, or for $t = 2$ with $B + 1$ a power of two. For $t = 2$, the eventual rank is $(\mathrm{odd}(B + 1) + 1)/2$, giving the identification with A003602.

For cycle lengths $t_1, \ldots, t_m$, the eventual rank is

$$\prod_{i=1}^{m} \frac{N_2(B, t_i) + 1}{2}.$$

Thus the dynamics depends on π only through its cycle type.

The argument is specific to a single generator. If maps associated with different permutations are composed, their cycle partitions need not agree, and the coordinatewise product decomposition is no longer available. The divisibility invariant $(B - 1) \mid F_\pi(x)$ survives such compositions, but the present semiconjugacy and rank formulas do not apply in general.

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